

# Probability and Statistics (MT-2005)

## Homework # 03(b)

### Joint Probability Distribution

1. Let  $X$  and  $Y$  be two discrete random variables representing the number of defective and non-defective products inspected in a factory. The joint probability mass function (PMF) is given below:

| $X \backslash Y$ | 0    | 1    | 2    |
|------------------|------|------|------|
| 0                | 0.10 | 0.15 | 0.20 |
| 1                | 0.05 | 0.25 | 0.10 |
| 2                | 0.05 | 0.05 | 0.05 |

- (a) Verify the Validity of the Joint PMF
  - (b) Compute the marginal probability mass functions  $g(X)$  and  $h(Y)$ .
  - (c) Compute Conditional Probabilities
    - (i) Find  $P(Y = 1|X = 1)$ .
    - (ii) Find  $P(X = 2|Y = 0)$ .
  - (d) Find the Expected Values i.e.,  $E[X]$ ,  $E[Y]$  and  $E[XY]$ .
  - (e) Find  $Var(X)$  and  $Var(Y)$ .
  - (f) Are  $X$  and  $Y$  independent? i.e.,  $f(X, Y) = g(X)h(Y)$ .
  - (g) Calculate  $Cov(X, Y) = E(X - E(X))(Y - E(Y)) = E(XY) - E(X)E(Y)$  and interpret the result.
  - (h) Find the correlation coefficient  $\rho = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$
2. Joint Probability Density Function (PDF) Let  $X$  and  $Y$  be two continuous random variables with the joint probability density function (PDF) given by:

$$f(x, y) = \begin{cases} c(6 - x - y), & 0 \leq x \leq 2, 2 \leq y \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Find the Constant  $c$  that ensures that  $f(x, y)$  is a valid joint PDF.
- (b) Compute the marginal PDFs  $g(X)$  and  $h(Y)$ .
- (c) Find the Conditional PDF
  - (i) Compute the conditional density function  $f(y|x)$ .
  - (ii) Compute the conditional density function  $f(x|y)$ .
- (d) Find the Expected Values i.e.,  $E[X]$ ,  $E[Y]$  and  $E[XY]$ .
- (e) Find  $Var(X)$  and  $Var(Y)$ .
- (f) Are  $X$  and  $Y$  independent? i.e.,  $f(X, Y) = g(X)h(Y)$ .
- (g) Calculate  $Cov(X, Y) = E(X - E(X))(Y - E(Y)) = E(XY) - E(X)E(Y)$  and interpret the result.
- (h) Find the correlation coefficient  $\rho = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$